

Noah Starkenburg

Software Engineer

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SUMMARY

Software engineer and B.S. Software Development graduate who builds full-stack web applications and backend services in C#, .NET, and SQL, with React and TypeScript on the front end and Go on the backend. Comfortable across REST APIs, relational and vector databases, testing, and cloud deployment, with production experience shipping real applications and code merged into open-source libraries.

EXPERIENCE

Automation Engineer (Intern), Byline Bank

April 2026 - Present

- Build internal tools and workflow automation on the Microsoft Power Platform with PowerShell, coding to specification and documenting each solution.
- Automated company-wide Active Directory lockout recovery with a PowerShell service and Power Automate flow, deflecting an estimated 80% of password-reset tickets and replacing a paid tool that cost about \$20,000 per year.

Software Developer, FamilyCore

September 2025 - April 2026

- Built and maintained a REST API in ASP.NET Core 8 with C#, including request validation, structured logging, per-session rate limiting, and security headers.
- Architected an agentic AI assistant that routes natural-language requests across 12 tools with a plan-and-execute loop and human-in-the-loop confirmation on all write actions.
- Built a RAG pipeline with Gemini embeddings, pgvector similarity search, and LLM reranking over a multi-schema PostgreSQL layer with Entity Framework Core and Hangfire background jobs.
- Grew the test suite to 622 integration and unit tests with xUnit, Moq, and Testcontainers, reaching 85% line coverage, and deployed to AWS Elastic Beanstalk with CloudWatch logging.

EDUCATION

Grand Canyon University, Bachelor of Science in Software Development

Graduated April 2026

GPA 3.63 / 4.0 · Honors College · 3rd Place, AWS Cloudathon

PROJECTS

KnowledgeMarket - Full-Stack eCommerce Marketplace *ASP.NET Core, React, SQL, Stripe, Docker*

- Built a full-stack course marketplace on a .NET 9 ASP.NET Core API serving 70+ REST endpoints across 7 domain modules, with a responsive React 19 and TypeScript single-page front end.
- Integrated the Stripe API for one-time and subscription payments with webhook-driven fulfillment, secured with JWT and CSRF, and covered by 250+ automated tests; deployed with Docker and GitHub Actions CI/CD.

Pulse Chat - Real-Time Chat System (in progress) *Go, WebSockets, Redis, PostgreSQL, Docker*

- Building a horizontally scalable, multi-room chat system in Go from first principles, with real-time WebSocket messaging, session auth, and Postgres-backed message history; Redis Pub/Sub scaling in progress.

Open Source - [alecthomas/kong](https://github.com/alecthomas/kong) *Go*

- Contributed a merged pull request to kong, an open-source Go command-line library with 3,000+ GitHub stars, adding support for running a command node with a Run() method without requiring a subcommand.

TECHNICAL SKILLS

Languages: C#, SQL, TypeScript, JavaScript, Go, Python, Java

Backend: ASP.NET Core, .NET, Entity Framework Core, REST APIs, Node.js / Express, JWT Auth

Frontend: React, Next.js, TypeScript, JavaScript, HTML / CSS

Databases & AI: PostgreSQL, SQL Server, Redis, pgvector, RAG pipelines, LLM integration, embeddings

DevOps & Tools: Docker, GitHub Actions, CI/CD, AWS, Git, xUnit, Testcontainers, Postman